

Introduction: (Done by Another Group Member)

For this project our goal is to design and build a wing box with maximum strength and minimum weight within the parameters of the design requirements. This is a very important project as the wing box relates directly to the planes ability to fly and carry cargo. It is also imperative to the safety to the plane as overly heavy or structurally weak/unstable components in the wing can cause catastrophic incidents to occur during flights leading to costly crashes injuries and deaths. This project is relevant in a range of engineering fields, most notably mechanical and aerospace. This is because the project heavily focuses on the different forces applied to the wing and how we can design it to handle these forces. It also focuses on the aerodynamics of the wing as along with withstanding force it also needs to provide lift and allow the plane to fly. The objective of this progress report is to highlight the progress we have made in the wing box project. It will include results and analysis from the various strength tests we have conducted, the methodology we have undertaken in conducting these tests, a materials investigation, a wing box geometry investigation and the initial wing box design we have come up with.

Problem Statement

When designing a plane, engineers aim to maximise the strength of a wing box whilst minimising weight. If no thought is to go into this then the wings may not be able to withstand the loads applied on them and the plane will become less efficient, losing range and fuel efficiency. The aim of this project is using aluminium materials to create the strongest wing box relative to its weight to identify the most effective designs.

Challenges to designing and building a wing box includes limits to our manufacturing capabilities, as each rib is made by hand and is therefore prone to human error, so each shape won't be perfect when under pressure. We are also unable to utilise certain designs such as an I-beam rib as they are too complicated to manufacture with the equipment available to us. Another consideration is that the aluminium sheets are unable to be cut, so the best way to minimise weight is to use a minimal amount aluminium sheets for the internal layout.

Wing boxes make up the structure of airplane wings and contribute between 49.9% to 61.3% on average of the planes total mass, so it is a vital component to its performance. When creating a wing box, engineers find the point where they can maximise its strength whilst making it as lightweight as possible. It is important to focus on strength as it is crucial to withstand shear, bending and torsional loads during flight to prevent structural failure, keeping the cargo of the plane safe. It also needs to be lightweight to maximise the planes efficiency, such as with fuel and range, meaning less money is spent on running the plane, which is better for airlines profits.

Methodology: (Done by Another Group Member)

Equipment:

- Universal testing machine – for stress test
- Tensile testing rig – to fit samples into universal testing machine for tensile test
- 3 point bending rig- to fit samples into universal testing machine for 3 point bending test
- Folding machine – for fabricating beams
- Pencil, ruler and paper – for drawing out designs
- Ruler and scribe – for marking bend lines on sheet metal
- Laptop and web access

Materials:

- Plastic tensile testing sample 90x10x3mm, Width x L_0 x Thickness
- Steel tensile testing sample 90x10x3mm, Width x L_0 x Thickness
- Aluminium tensile testing sample 90x10x3mm, Width x L_0 x Thickness
- Aluminium sheet 200x100mm

Activity 1, Tensile Test:

Using the universal testing machine each sample was placed in the grips where a constant pressure was applied until failure where upon the sample was taken out and replaced by the other until all three samples were tested. The results of force, displacement and time were collected for each sample.

figure 1



Activity 2, Beam design:

Using the bending stress formula, it was found that to resist greater strength the beam must have a higher moment of inertia along the neutral axis. Taking this information, different cross-sectional profiles were researched to investigate. The profile of a top hat batten (figure 2) was found and decided upon to investigate alongside a basic square cross section.

Figure 2. (ref 3)

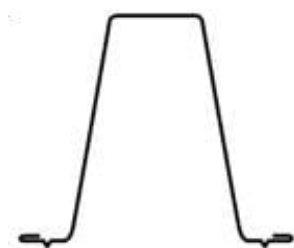


Figure 3. (ref 2)

$$\sigma_b = \frac{My}{I}$$

σ_b – Bending stress

M – Calculated bending moment

y – Vertical distance away from the neutral axis

I – Moment of inertia around the neutral axis

After this the appropriate dimensions were calculated, and a final schematic of each beam design was sketched.

Figure 4

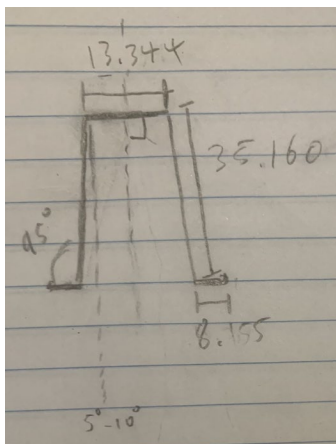
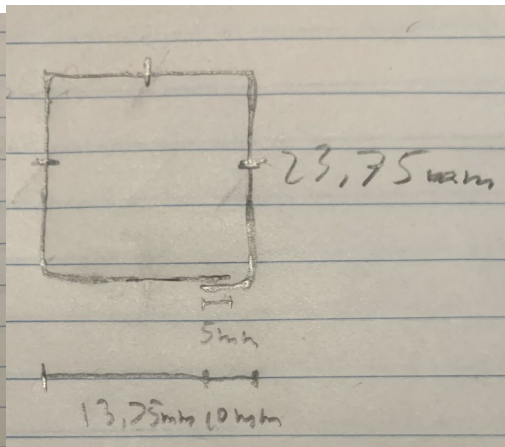


figure 5



After both schematics were sketched and checked by a lab technician the fold lines were scribed into the sheet metal, and the designs were bent into shape accordingly.

Activity 3, Bending test:

Both beams were placed into the universal testing machines 3 point bending rig one at a time and a constant force was applied until the beam failed. The displacement, force and time was measured for both samples during the test.

Figure 6.



Wing Box Materials Investigation: (Done by Another Group Member)

Materials Description:

In the tensile test three materials were tested steel, aluminium and plastic as described;

Each sample were shaped as seen in figure 6.

Figure 7.

Steel sample

Dimensions- 90x10x3mm, Width x L_0 x Thickness

Weight- 44g

Aluminium sample

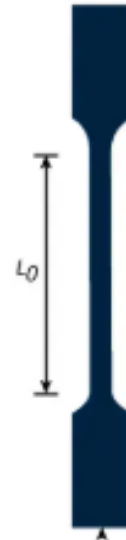
Dimensions- 90x10x3mm, Width x L_0 x Thickness

Weight- 15g

Plastic sample

Dimensions- 90x10x3mm, Width x L_0 x Thickness

Weight- 3g



Test discussion:

This test setup itself was ideal for operating with a high level of precision and removing errors. This precision is evident in the two decimal places for displacement and four decimal places for force seen on the display on the right of figure 7. To stay consistent each sample had the same dimensions. The testing procedure consisted to simply placing the sample into the tensile testing rig and applying a constant force until failure. The only downfall of this procedure was the lack of samples tested for each material with only one sample tested instead of multiple to decrease random sampling errors through averaging the values.

Figure 8.



Figure 9.

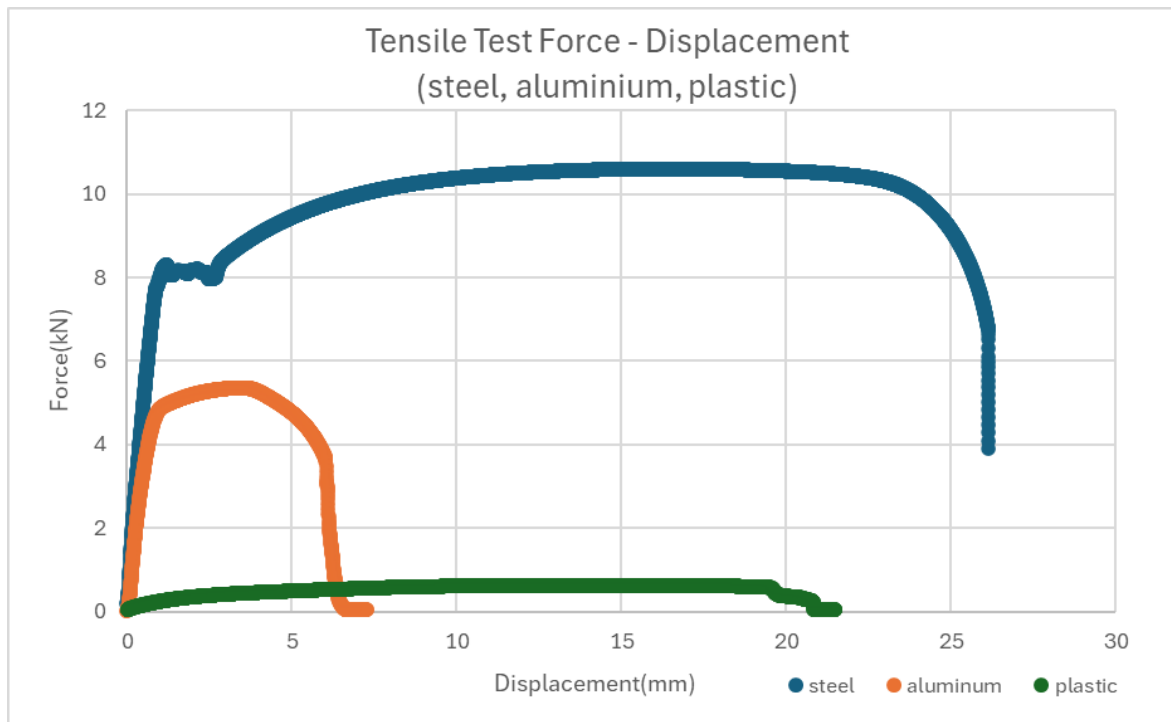
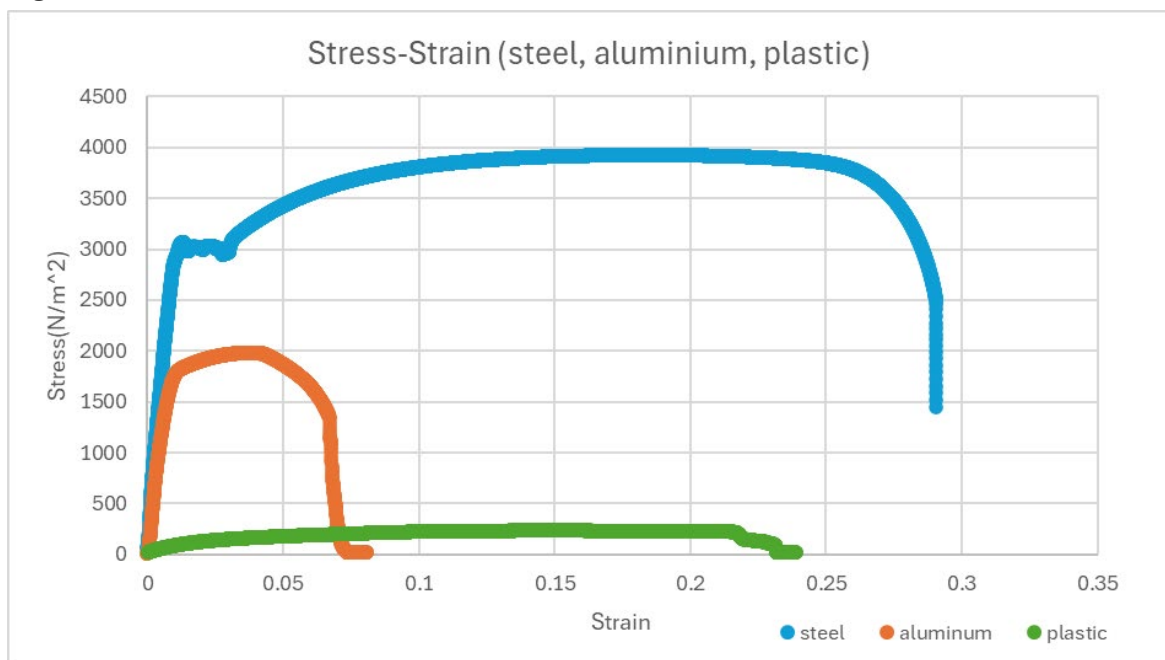


Figure 10.



Results analysis:

Steel

Tensile strength: 3924.67 N/m²

According to figure 10 it can clearly be seen that steel outstrips aluminium and plastic in tensile and yield strength. Steel also has a high young's modulus making it a very stiff

material and less prone to bending. Its high yield strength also means it can withstand more force before permanent deformation but once it does start to deform it does so in a more sudden way than aluminium and plastic making it more dangerous to use if it could be subjected to forces greater than its yield strength.

Plastic

Tensile strength: 233.04 N/m²

As seen in figure 10 plastic has the smallest young's modulus making it the most able to bend without deformation out of the three materials. However, the combination of having such a relatively small tensile and yield strength makes it less suitable to use cases where it will be subjected to large forces. The only other property of interest of this material is its incredibly light weight compared to the other materials, only 3g next to steel(44g) and aluminium(15g) with the same volume.

Aluminium

Tensile strength: 1982.37 N/m²

Aluminium takes the middle ground of steel and plastic in every main mechanical property. It deforms less violently than steel but also has a higher young's modulus than plastic. Aluminium has more tensile and yield strength than plastic but less than steel. Ultimately aluminium is secondary to steel in terms of resistance to forces applied to it and its only benefit over steel is a decreased weight of 15g instead of steels 44g.

Summary

Tensile strength: Steel > Aluminium > Plastic

Yield strength: Steel > Aluminium > Plastic

Young's modulus: Steel > Aluminium > Plastic

Stiffness: Steel > Aluminium > Plastic

Flexibility: Plastic > Aluminium > Steel

Density: Steel > Aluminium > Plastic

Comparison for wing box use case:

For a wing box there are two key mechanical elements to consider when selecting which material to use, strength and weight. The wing box must be able to resist the intense and inconsistent forces it will be subjected to in flight. It must also weigh as little as possible since each kilogram used to construct the plane corresponds to greater fuel usage and hence money spent.

Hence the final material chosen should be a balance between weight and strength. When referring to the summary of the results analysis it can be clearly seen that in every way aluminium falls between steel and plastic. Steel is very strong but is almost 3 times as dense as aluminium. Plastic is flexible and extremely lightweight but has a tensile strength

8.5 times less than aluminium. Therefore, while it is almost 2 times weaker than steel, aluminium is our chosen material out of the three due to its significantly lower density.

Wing Box Geometry Investigation: (Done by Another Group Member)

Wing Box Design

Design	Weight	Strength	Manufacturability
Top hats (2)	1	2	3
Squares (3)	1	1	2
Alternating top hats (3)	2	3	2
Squares and top hats	3	4	1
Two top hats			

Design 1

The first design is two aluminium top hat cross section beams placed next to each other. each top hat is approximately 35mm in height with a 35.8 mm base and 250mm length. There is also a 100mm by 250mm aluminium sheet with 2mm thickness placed above and below the three beams. The configuration of these top hats is shown in the figure below (Figure 1). Each beam weight 50 grams bringing the total weight (ignoring screws/pop rivets) to 200 grams.

Design 2

The second design is three aluminium square cross section beams placed next to each other. Each beam is approximately 23mm by 23mm and are all once again 250mm in length with two 100mm by 250mm aluminium sheets that are 2mm thick placed above and below the beams. The configuration for design 2 is shown below. The weight for design 2 is also 250 grams (ignoring screws and pop rivets)

Design 3

The third design is similar to design 1 in that it utilises the same top hat shaped beams with the same specifications, the difference is that design 3 has three beams instead of two and it utilises a different configuration. While the first design had all the beams facing the same way the third design has them alternating between upright and upside down. this allows for three beams to fit between the sheets instead of just two. The weight of this design is 250 grams and the aluminium sheets are 100mm by 250mm. The configuration is found below.

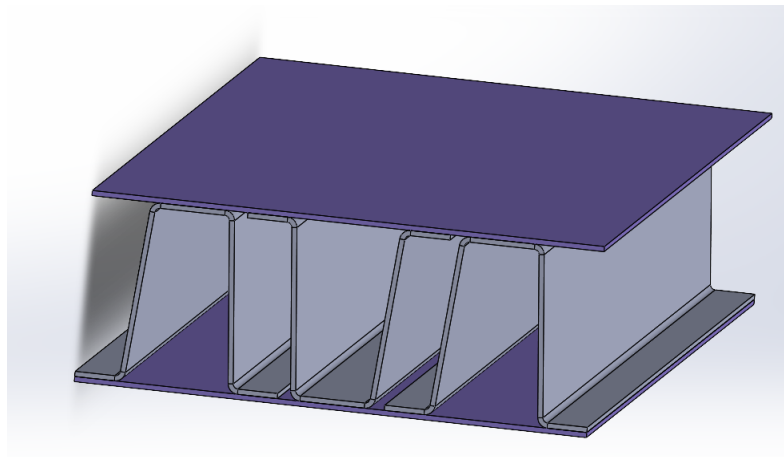


Figure 1

Design 4

Design four is a combination of designs 1 and 2 meaning it will use both top hats and squares.

Alternating Rib Orientation

Initial Design

For this wing box design, we will use three top hat ribs with the dimensions in *figure 3*. They will be colinear and sandwiched between two 250 mm x 100 mm x 2 mm aluminium sheets. These components will be jointed using 2.5 mm rivets with their centre located 3.16mm from the edge of each rivets foot on both ends. This totals to 4 rivets used per rib, or 12 for the whole wing box.

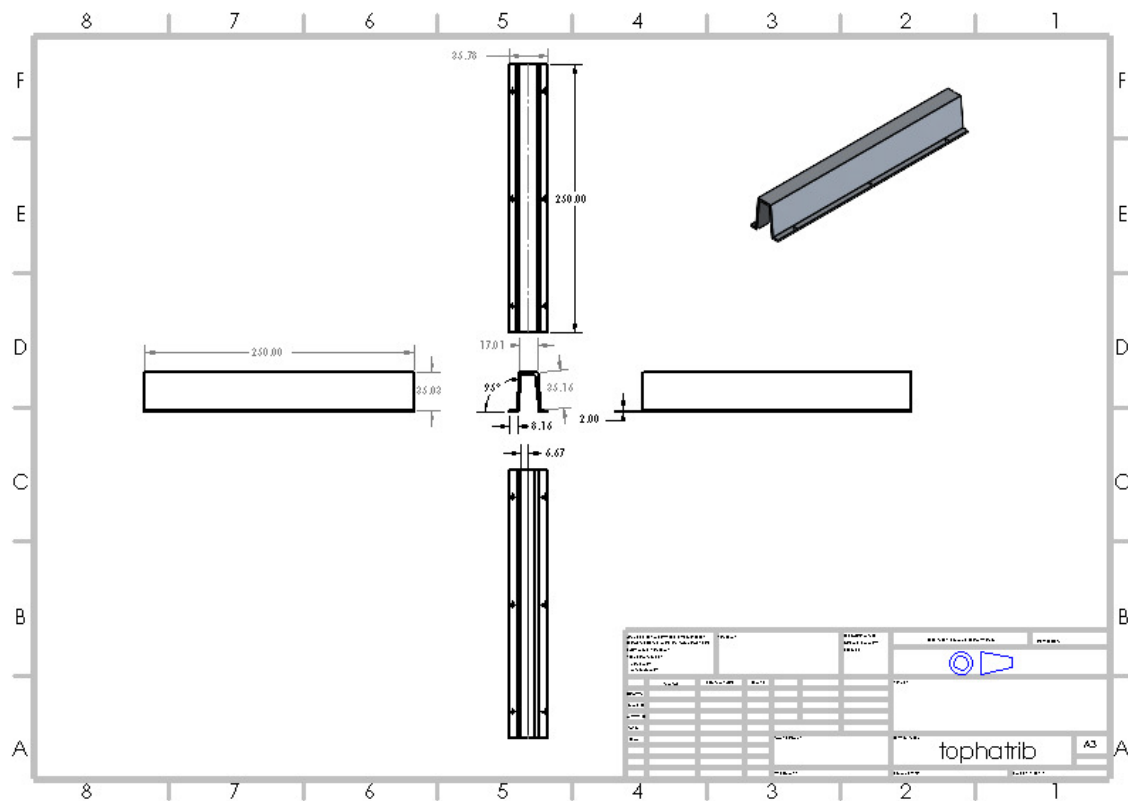


Figure 2

Design Considerations

The placement of the rivets changed from 4 per rib, with one situated at either end of both feet to distribute the load among the fasteners, reducing the risk of failure at the connection points. Having rivets closer to one another also reduces unsupported areas in the skin, which in turn improves the wing box's buckling resistance. From team discussions and analysis of the design parameters, it was decided that an excessive use of rivets reduced its manufacturability, limited by construction time available as well as the extra weight added by the rivets to the wing box. We landed on the current rivet layout, *figure 4,5,6*, as the point in which we believe the optimal trade-off occurred.

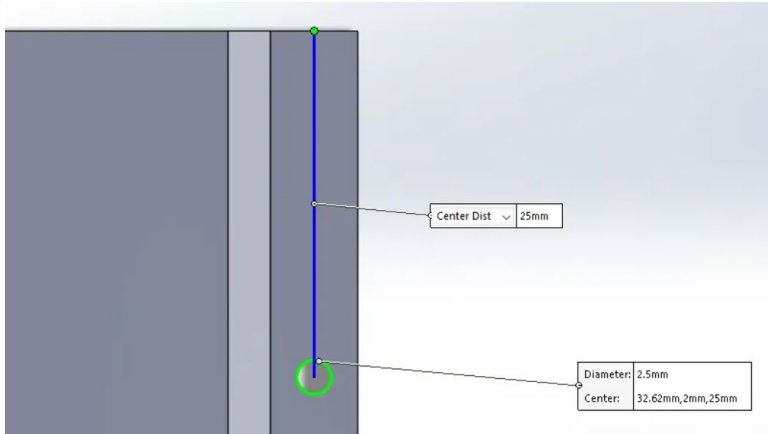


Figure 5

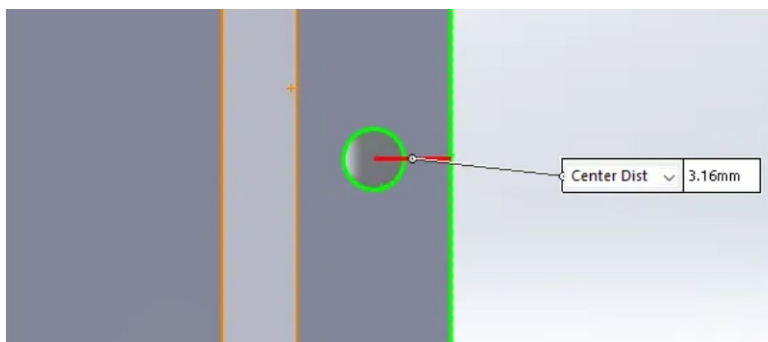


Figure 5

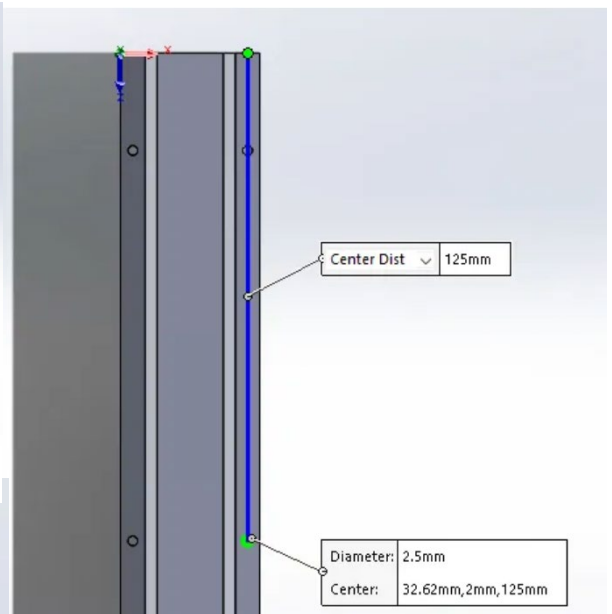


Figure 5

Another iteration discussed by our team was to not have our ribs alternating orientation, and instead all sitting flat on the lower skin. When designing this iteration, it was found that the limit of the area of the lower skin didn't allow for this to be viable. The width of one rib is 35.78mm, *figure 7*, so with 3 of parts sitting colinear they wouldn't have fit inside the bottom skin's constraints. This led to the current design of the alternating orientation ribs as when the top of the rib has a smaller width when aligned, they fit inside the design limits. Additionally, looking at similar structures in studies we can better understand the best way to increase the wing box's bending stiffness. With our assembly having thin skin, adding stiffening ribs, such as without top hat design it will "considerably enhance" (Harvey & Virgin 2018) the lateral bending stiffness through disproportionately the second moment of inertia whilst not adding relatively more mass. Therefore, if our design aligns with Harvey and Virgin's 2018 conclusion on the effects of stiffeners, we can create a wing box that deflects less under an identical load.

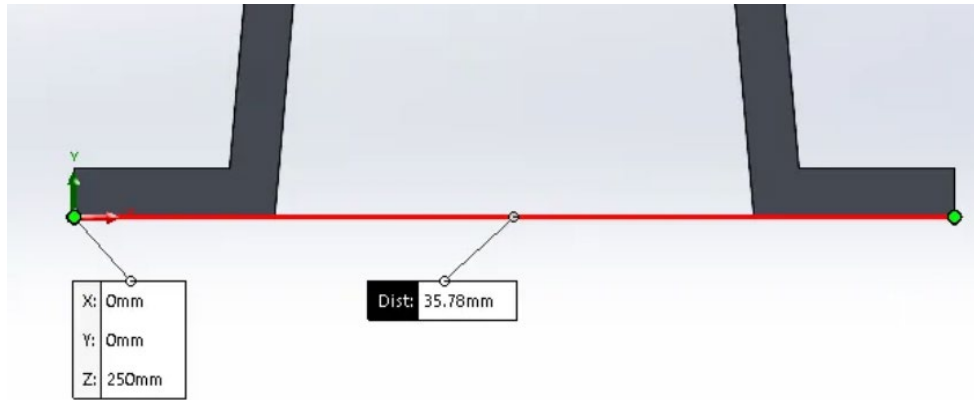


Figure 7

With the wing box constructed using identical aluminium sheets, this design reduces the weight as much as possible while still resting high loads, using a minimal number of ribs as the upper and lower skins locked in as design constraints. When testing different designs for the rib, the top hat had the highest load capacity as seen in *figure 8*, withstanding a maximum load of 1347N. Boeing's patent for 'Composite aircraft structures with hat stiffener's' (Lee et al. 2006) suggests that top hat rib designs offer an enhanced torsional rigidity, buckling resistance and bending stiffness. This aligns with our findings from testing our ribs designs in a 3 point bend test. As the only material available for use was aluminium, when it came to optimising the wing box's structural stability it was essential to also have a rib that not only had a high load capacity but also had a large displacement. Whilst the top hat rib had a maximum displacement of 4.1292mm before the part fractured, the square rib had a greater displacement at 7.7059mm before fracturing. However, it should be noted this occurred at a far lighter load force of 830.5N. Even though the square shaped rib displaced further before failure, its low load capacity makes the top hat rib more ideal. Further supporting the requirement for a rib with stiffer geometry to better improve the failure mode and strength of the wing box by more effectively distributing load as well as delaying instability.

Final Design

The final design will use 3 top hat ribs sitting colinear with an alternating orientation, which can be seen in *Figure 5*. Both the ribs and skins are created using 250 mm x 100 mm x 2 mm aluminium sheets. The dimension of each rib is shown in *figure 3*, remaining the same as in the Specimen 1 to 2

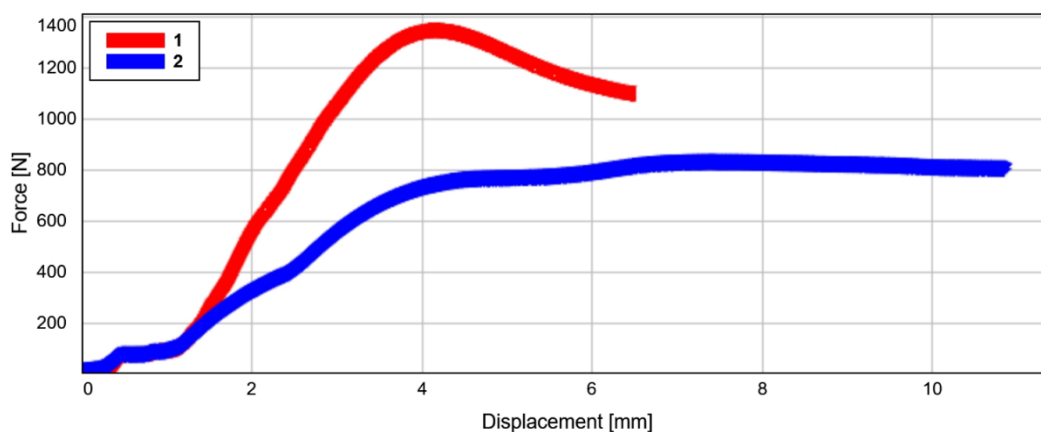


Figure 6

initial design. The ribs all have identical holes and hole locations. There are 6 holes on each rib with 3 on either foot or all being in alignment. The diameter of each hole is $\varnothing 2.5$ mm. Each hole is colinear along the length of the rib, with the hole centre offset by 3.16 mm. In relation to *figure 9*, 2 holes are 25 mm from the front edge, 2 holes are 25 mm from the rear edge, and the final 2 125 mm from the front edge which is the centre of the rib. The bottom skin has 12 holes each with a diameter $\varnothing 2.5$ mm. The locations of these can be found in *figure 10*. The top skin has 6 holes with a diameter $\varnothing 2.5$ mm with the locations from *figure 11*. When the wing box is assembled in accordance with *figure 12*, these holes are aligned and are where the rivets will be inserted. In total there is 18 2.5 mm rivets. This new design will be a similar weight with the same number of aluminium sheets used, although it differs slightly as there are more holes therefore lessening its mass, but also more rivets, increasing its mass. Given that the material used throughout the wing box is aluminium 3003 alloy, its mass is approximately 675.46 grams, as shown with the information in *figure 13*. Additionally, when using *figure 14*, we see that this wing box design has a cross sectional area of 1024.55 mm². Using information from these figures we can calculate the parameters such as bending strength, failure load, bending stiffness and the strength to weight ratio.

Bending stiffness: $\Sigma = EI = 69000 \times 259333.65 = 1.789 \times 10^{10}$

Deflection at the middle of the wing box under a load P creates the graph for the relation

between the applied load and the midspan deflection: $\Delta_{mid} = \frac{PL^3}{48EI} = \frac{P \times 250^3}{48 \times (1.789 \times 10^{10})}$

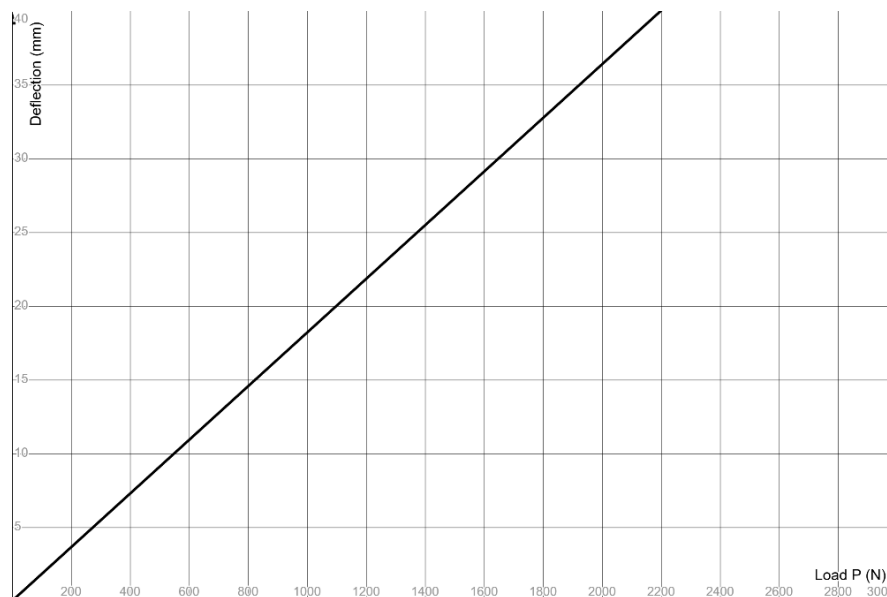


Figure 10

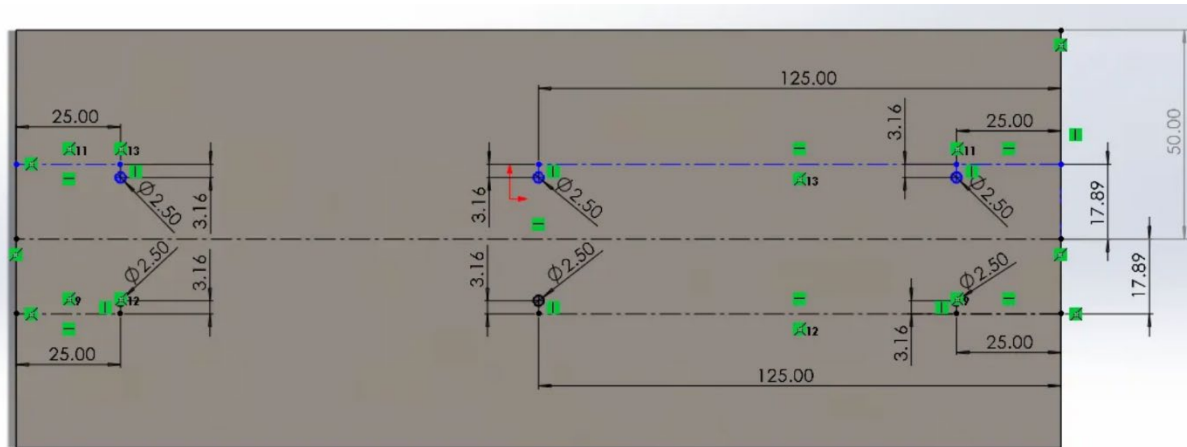


Figure 11

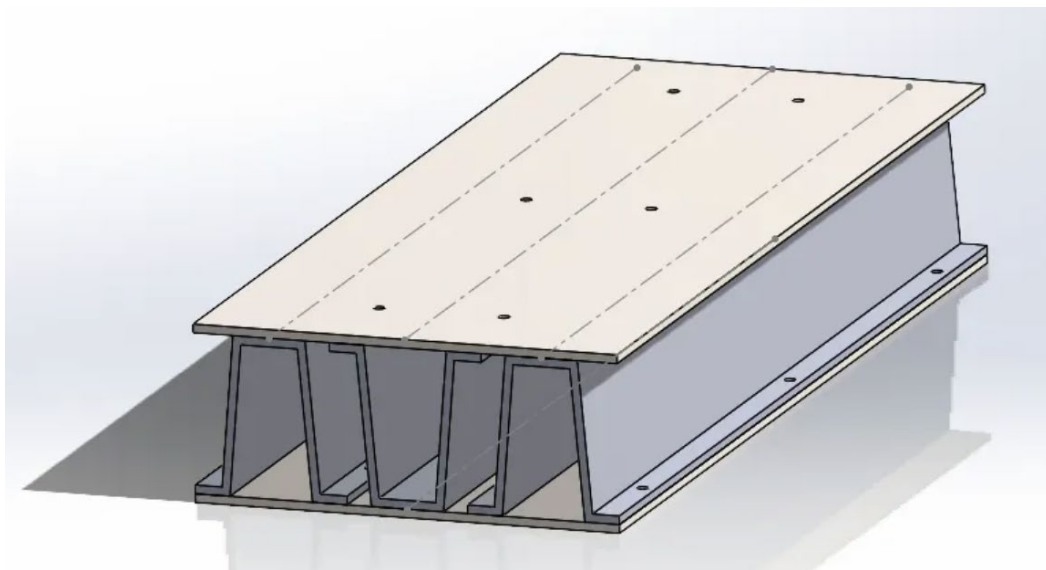


Figure 12

Mass properties of wingbox - Copy
Configuration: Default
Coordinate system: -- default --

Mass = 675.46 grams
Volume = 250171.11 cubic millimeters
Surface area = 258550.88 square millimeters

Center of mass: (millimeters)
X = -36.3
Y = -2.57
Z = 47.64

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)
Taken at the center of mass.

Ix = (0, 0, 1)	Px = 710900.82
Iy = (1, -0.01, 0)	Py = 3688181.59
Iz = (0.01, 1, 0)	Pz = 4060437.77

Moments of inertia: (grams * square millimeters)
Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

Lxx = 3688221.13	Lxy = -3836.32	Lxz = 2.41
Lyx = -3836.32	Lyx = 4060398.23	Lyz = 0.85
Lzx = 2.41	Lzy = 0.85	Lzz = 710900.82

Moments of inertia: (grams * square millimeters)
Taken at the output coordinate system. (Using positive tensor notation.)

Ixx = 5225499.07	Ixy = 59304.12	Ixz = 1168152.89
Iyx = 59304.12	Iyy = 6483455.71	Iyz = 82849.13
Izx = 1168152.89	Izy = 82849.13	Izz = 1605636.68

Figure 13

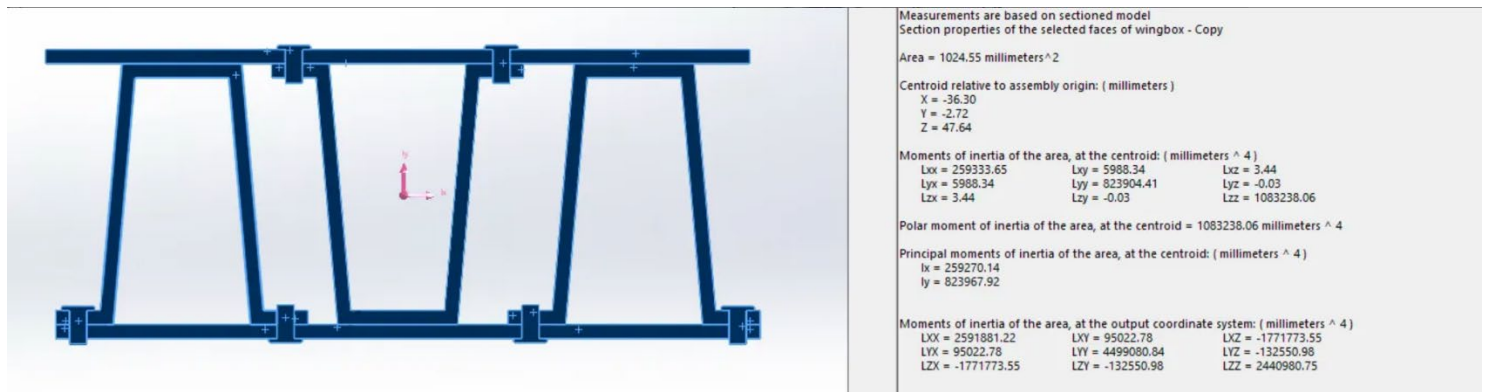


Figure 14

Analysis of results

The design featuring an alternating rib orientation is performs optimally in calculations that indicate the wing box's ability to withstand an applied load before fracturing.

It should be noted that all values used in this section have been obtained from the CAD design and simulations located in the *wing box design* section under the alternating rib orientation design. 3-point bend test simulation figures are shown using a 1000 N load applied to the wing box with the material of all wing box components being aluminium 3003 alloy and the material properties sourced from the SolidWorks materials library. These simulations were conducted with the wing box's being as faithful to actual conditions as possible. Similar validation testing is standard in industry component design to prove calculations for analysis of the wing box performance. Given that our physical wing box is created like the CAD design, its performance is expected to closely match that of the simulation. Through using a 3-point bend test to test we will be able to confirm the alternating rib orientation's ability to delay buckling or yielding in a wing box.

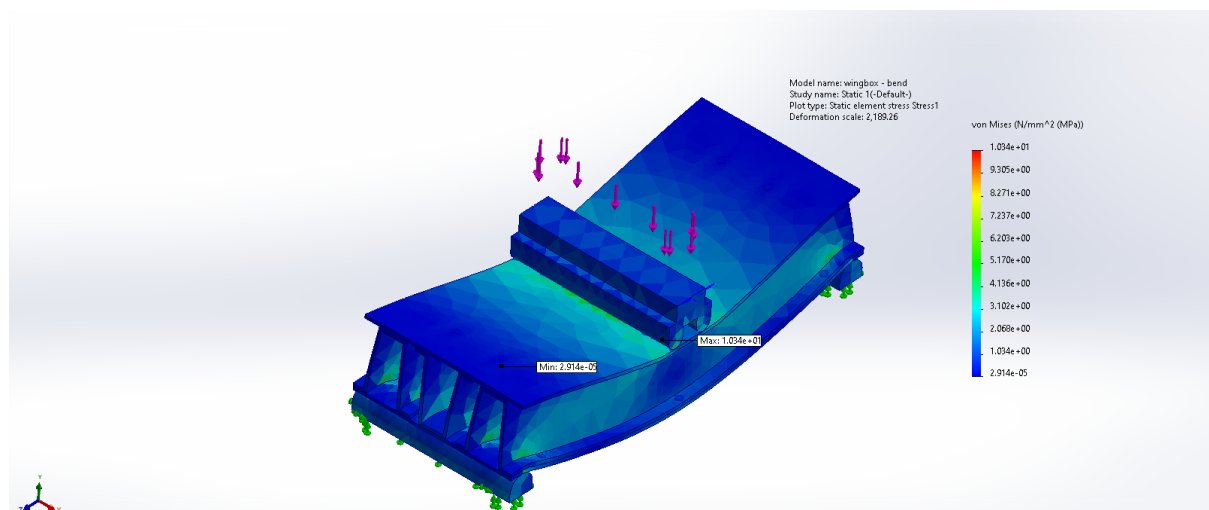
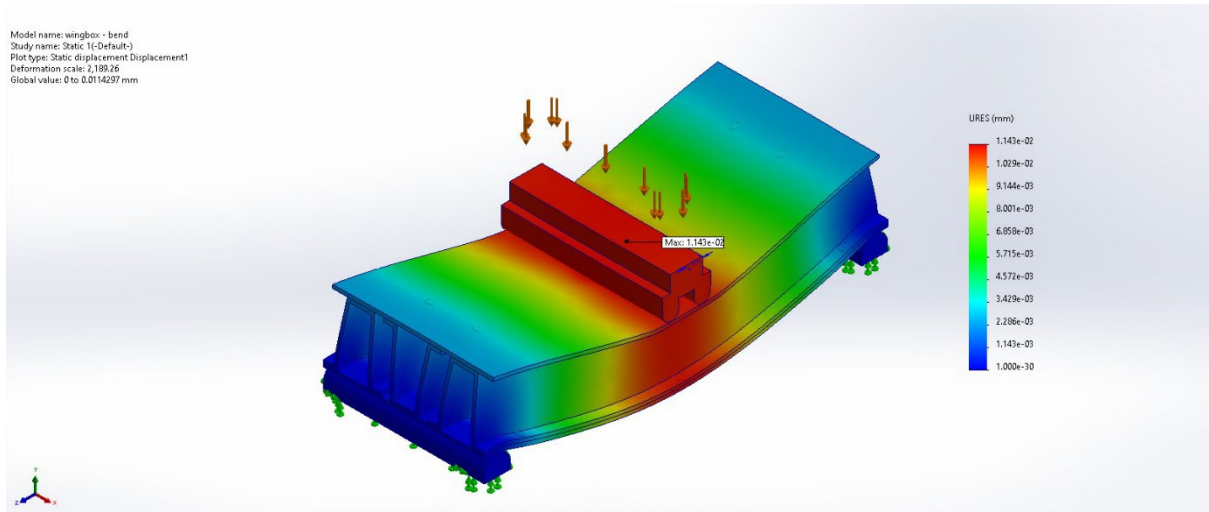
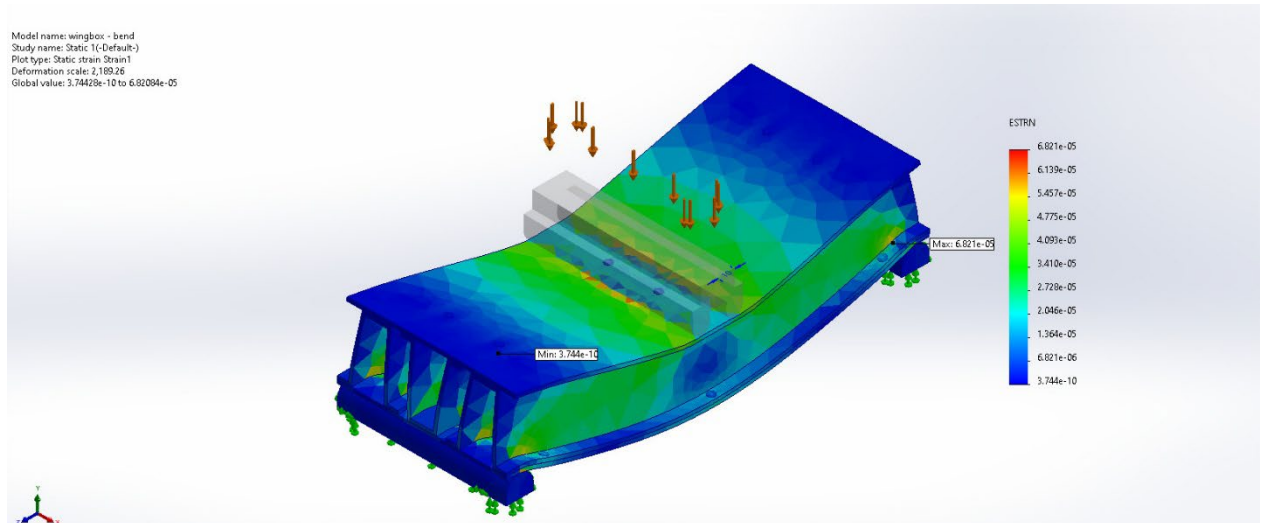


Figure 15 (stress simulation)

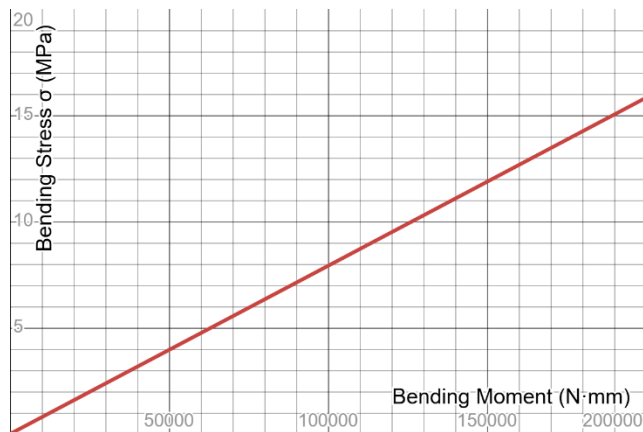


(Displacement Simulation)



(Strain Simulation)

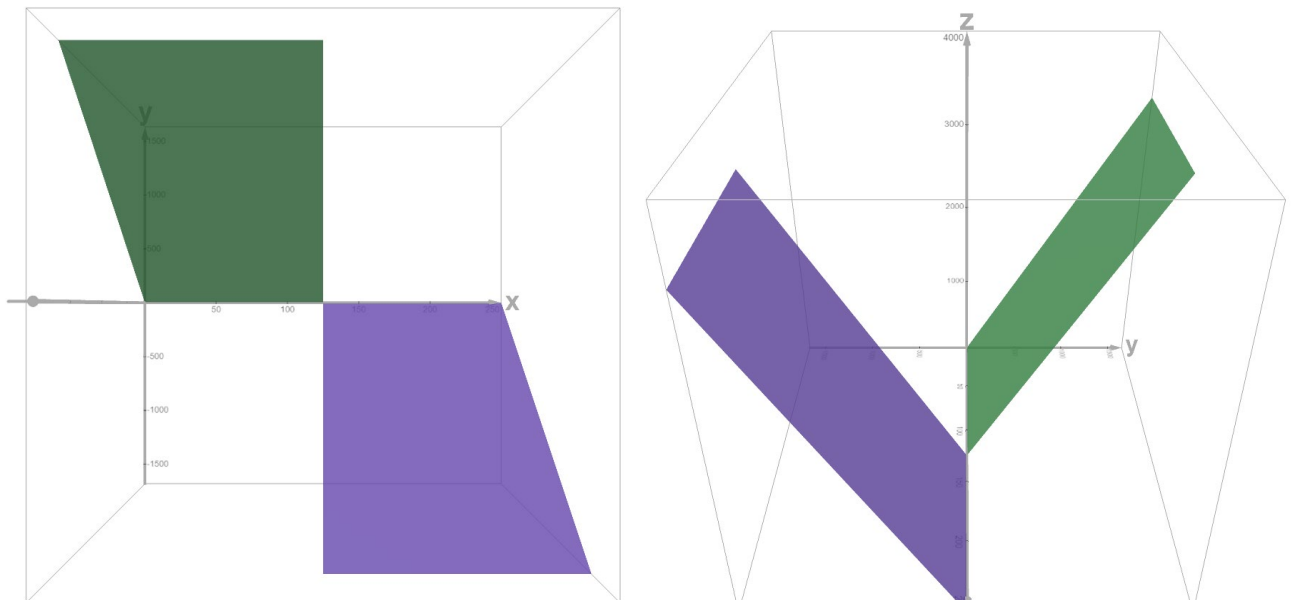
The alternating rib orientation design also has a greater bending stiffness, $\Sigma = EI = 69000 \times 259333.65 = 1.789 \times 10^{10}$, where the assumed material used is aluminium 3003 alloy. This is due to the additional rib increasing the wingbox's cross-sectional area. The greater bending stiffness directly reduces deflection as shown by the equation, $\Delta_{mid} = \frac{PL^3}{48EI}$, and since bending stiffness occurs in the denominator, increasing EI leads to a decrease in in deflection for an identical applied load. The relationship between bending stress σ and the bending moment M is found from the equation $\sigma = \frac{My}{I} \Rightarrow \sigma = \frac{M \times 20.52}{259333.65}$ and can be visualised through the following graph.



Load distribution is governed by the equilibrium equations for shear, reactions and moments.

The shear force distribution can be found from $V(x) = \begin{cases} +\frac{P}{2}, & 0 < x < 125 \\ -\frac{P}{2}, & 125 < x < 250 \end{cases}$, and is graphed

below for each point along the airfoil.

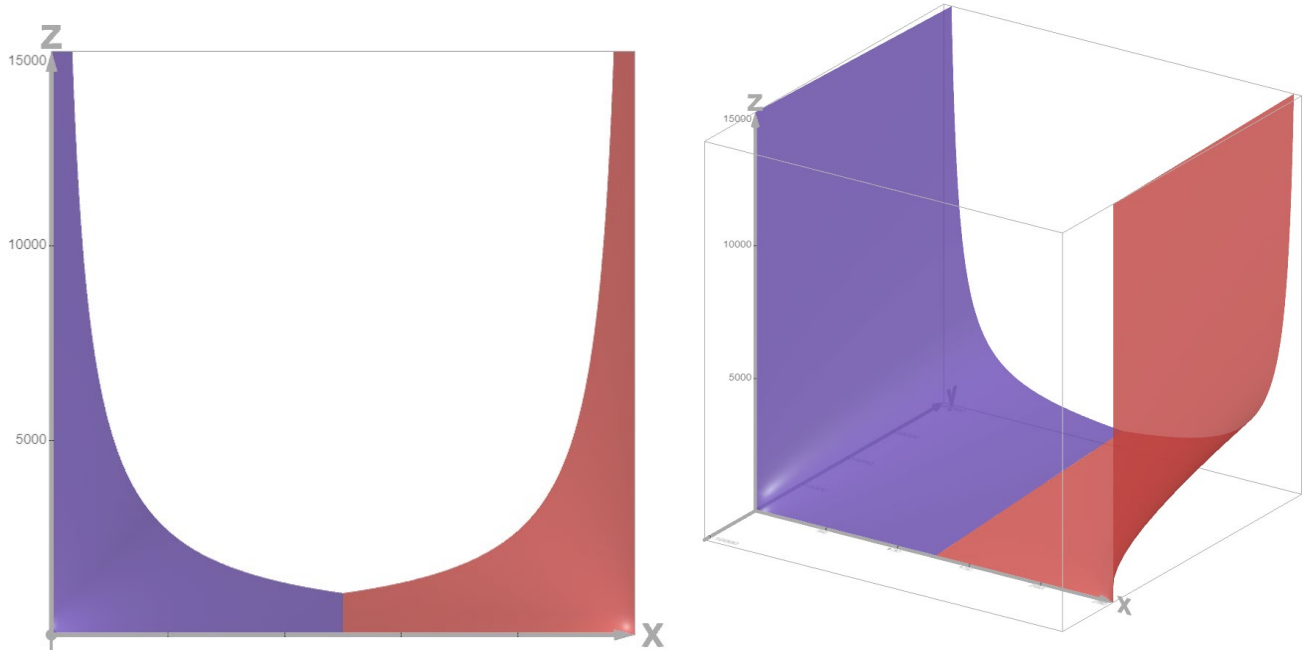


Where the x-axis is the location along the length of the wing box in mm, the y-axis is the shear force $V(x)$ in newtons and the z-axis is the load P in newtons.

Support reactions are equal as $R_A = R_B = \frac{P}{2}$, which is supported through the graph above depicting the shear force distribution, as each length is equal and opposite.

The bending moment distribution can be found from $M_{(x)} = \begin{cases} \frac{P}{2}x, 0 \leq x \leq \frac{L}{2} \\ \frac{P}{2}(L-x), \frac{L}{2} < x \leq L \end{cases}$, where the maximum moment is the centre and can be found in the equation given by the equation $M_{max} = \frac{PL}{4}$, where the x-axis is the position along the wingbox in mm, the z-axis is the shear bending moment in N · mm and the y-axis is the applied load at the midpoint in newtons.

The graph on the left shows exponential increase in the bending moment towards the edges of the wingbox when a load is applied. This demonstrates how load distribution is critical to reduce local stress at the end points which better prevents local failure in the skin or ribs.



When CAD modelling and completing calculations for bending stress and deflection, there was an uncertainty around which aluminium alloy will be used in the physical design. Due to differing yield strengths, elastic moduli and densities, using a different alloys mechanical property would return inaccurate results to what would be found in the wing box tests. While the project specification required the use of aluminium sheets, it didn't specify the alloy. For purposes of simulations and analysis of the wing box, the assumption was made for the aluminium to be 3003 alloy as a general option. Its values were taken from the SolidWorks materials library to ensure constant values across all areas where that information is needed. Although it should be noted that if a different alloy is to be used in the construction of the wing box than the calculations made in the *wing box design* and *analysis of results* sections won't be faithful to reality due to differing stiffnesses and yield strength.

Another area of challenge in the design process was the lack of information on the rivets. The project specification had no mention of the dimensions, type, material or spacings for the rivets. Due to the lack of information provided, the design created in the *wing box model* section may produce different results than what was found in the studies. This effects integral components to the final design decision such as the weight, cross section, stress & displacement studies, and the deflection. For purposes of keeping the wing box as faithful to reality as possible with

the limit information provided, we assumed the rivets were AS B118 Flat – 0.094 x 0.188, which can be found in the SolidWorks toolbox and whose material was the same aluminium 3003 alloy as the rest of the assembly. This ensured a constant material across the wing box.

Conclusion: (Done by Another Group Member)

Bibliography

INTRODUCTION	
	Stramit Building Products <i>Product Technical Manual: StraMit Top Hats & Battens Capacity Tables</i> , viewed < https://www.stramit.com.au/assets/Technical-Resource-Documents/6fec0f3837/Top-Hats-Battens-Capacity-Tables-Product-Technical-Manual.pdf >.
	Spinney, J 2025, “Stress formula and equation in beam calculations: understanding bending, shear, normal stress, and tensile strength,” <i>StruCalc</i> , viewed < https://strucalc.com/blog/learning/bending-shear-and-normal-stress-formulas/ >.
PROBLEM STATEMENT	
	Chambers, MC, Ardema, MD, Patron, AP, Hahn, AS, Miura, H & Moore, MD 1996, “Analytical fuselage and wing weight estimation of transport aircraft,” <i>NASA Technical Reports Server (NTRS)</i> , viewed < https://ntrs.nasa.gov/citations/19960025262 >.
WING BOX GEOMETRY INVESTIGATION	
	Bisagni, C, Vescovini, R & Davila, C 2010, “Assessment of the Damage Tolerance of Post-Buckled Hat-Stiffened Panels using Single Stringer Specimens,” <i>AIAA</i> .
Wing Box Design	
	Harvey, P Scott, Jr & Virgin, L 2018, “Effect of Stiffener Geometry on the Response of Grid-Stiffened Panels,” <i>Journal of Engineering Mechanics</i> .
	Lee, SS, Viisoreanu, A, Koch, WJ & Co, B 2006, “US7871040B2 - Composite aircraft structures with hat stiffeners - Google Patents,” viewed < https://patents.google.com/patent/US7871040B2/en#:~:text=,manufacturing%20cost%20compared%20with%20%E2%80%9Copen%E2%80%9D >.

	Korba, P, Al-Rabeei, S, Hovanec, M, Sekelová, I & Kale, U 2024, “Structural design and material comparison for aircraft wing box beam panel,” <i>Heliyon</i> , vol. 10, no. 5, p. e27403.
INTRODUCTION	
	Stramit Building Products <i>Product Technical Manual: StraMit Top Hats & Battens Capacity Tables</i> , viewed < https://www.stramit.com.au/assets/Technical-Resource-Documents/6fec0f3837/Top-Hats-Battens-Capacity-Tables-Product-Technical-Manual.pdf >.
	Spinney, J 2025, “Stress formula and equation in beam calculations: understanding bending, shear, normal stress, and tensile strength,” <i>StruCalc</i> , viewed < https://strucalc.com/blog/learning/bending-shear-and-normal-stress-formulas/ >.
	Stramit Building Products <i>Product Technical Manual: StraMit Top Hats & Battens Capacity Tables</i> , viewed < https://www.stramit.com.au/assets/Technical-Resource-Documents/6fec0f3837/Top-Hats-Battens-Capacity-Tables-Product-Technical-Manual.pdf >.
PROBLEM STATEMENT	
	Chambers, MC, Ardema, MD, Patron, AP, Hahn, AS, Miura, H & Moore, MD 1996, “Analytical fuselage and wing weight estimation of transport aircraft,” <i>NASA Technical Reports Server (NTRS)</i> , viewed < https://ntrs.nasa.gov/citations/19960025262 >.
WING BOX GEOMETRY INVESTIGATION	
	Bisagni, C, Vescovini, R & Davila, C 2010, “Assessment of the Damage Tolerance of Post-Buckled Hat-Stiffened Panels using Single Stringer Specimens,” <i>AIAA</i> .
Wing Box Design	

	Harvey, P Scott, Jr & Virgin, L 2018, “Effect of Stiffener Geometry on the Response of Grid-Stiffened Panels,” <i>Jounral of Engineering Mechanics</i> .
	Lee, SS, Viisoreanu, A, Koch, WJ & Co, B 2006, “US7871040B2 - Composite aircraft structures with hat stiffeners - Google Patents,” viewed < https://patents.google.com/patent/US7871040B2/en#:~:text=,manufacturing%20cost%20compared%20with%20%E2%80%9Copen%E2%80%9D >.
	Korba, P, Al-Rabeei, S, Hovanec, M, Sekelová, I & Kale, U 2024, “Structural design and material comparison for aircraft wing box beam panel,” <i>Heliyon</i> , vol. 10, no. 5, p. e27403.

Section	Completion by:
Introduction	
Problem Statement	Hugh
Methodology	
Wing Box Materials Investigation	
Wing Box Geometry Investigation	
Wing Box Design	Hugh
Analysis of Results	Hugh
Conclusion	